

Functional & Performance Testing Template

Model Performance Test

Date	22 June 2026
Team ID	Not Given
Project Name	Rising Waters
Maximum Marks	

Team Members	Skill Wallet Team ID's	Roll Numbers (College: ACET)
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Test Scenarios & Results

The Rising Waters flood prediction system underwent comprehensive testing to verify its functionality, reliability, and overall performance. Functional testing was conducted on all major modules, including user authentication, weather data input, data preprocessing, machine learning model integration, flood prediction, result visualization, and prediction history management. Each module was tested using both valid and invalid inputs to ensure the system handled different scenarios accurately and provided appropriate responses.

Performance testing evaluated the application's response time, stability, and ability to process prediction requests efficiently. The machine learning model was tested using historical weather data to validate its prediction accuracy, while the Flask web application was assessed for smooth interaction between the frontend and backend. Integration testing confirmed successful communication between the user interface, database, preprocessing module, and trained machine learning model. User acceptance testing ensured that the application was easy to use and met the requirements of disaster management authorities and other stakeholders.

The testing results demonstrated that the system performed reliably, generated accurate flood predictions, responded within acceptable time limits, and successfully met the functional and performance requirements. Overall, the Rising Waters application proved to be stable, efficient, and ready for deployment in real-world flood monitoring and early warning scenarios.

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	User Login & Authentication	Enter valid and invalid login credentials	Valid users can log in successfully; invalid users receive an error message	As Expected	Pass
FT-02	Weather API Data Collection	Connect to Weather API and fetch live weather data	Weather data is retrieved and displayed correctly	As Expected	Pass
FT-03	River Water Level Monitoring	Fetch river water level data from sensors/API	Latest river water level is displayed accurately	As Expected	Pass
FT-04	AI Flood Prediction	Provide rainfall, weather, and river level data to the ML model	System predicts flood risk as Low, Medium, or High	As Expected	Pass
PT-05	Alert Notification System	Simulate high flood risk conditions	SMS, Email, and Mobile alerts are sent to users	As Expected	Pass
PT-06	Dashboard Visualization	Open dashboard after prediction	Dashboard displays weather, flood risk map, and prediction reports	As Expected	Pass